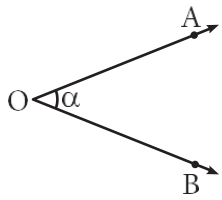
Clasificación de los Ángulos

• Marco teórico

Un ángulo es una figura geométrica formada por la unión de dos rayos mediante un origen común, llamado vértice del ángulo. Los rayos son los lados del ángulo.



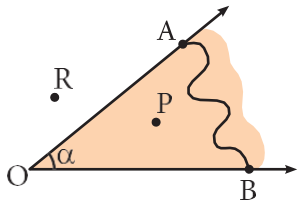
Elementos

- Lados : _____
- Vértice : _____

Notación:

- ∠AOB : ángulo AOB
- m∠AOB: medida del ángulo AOB

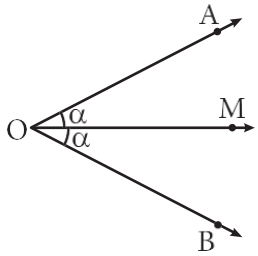
Observación:



Donde:

«P»: pertenece a la región interior del ángulo.
«R»: pertenece a the región exterior del ángulo.

La bisectriz es el rayo que biseca al ángulo



$\overrightarrow{OM}$: **Bisectriz del ∠AOB**

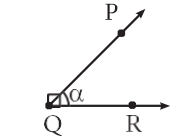
Si $\overrightarrow{OM}$ es bisectriz del ∠AOB, se cumple:

m∠AOM = m∠MOB = **α**

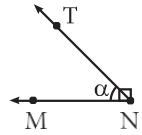
CLASIFICACIÓN DE LOS ÁNGULOS

Según sus medidas:

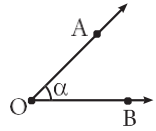
a) Ángulo agudo



$0° < \alpha < 90°$

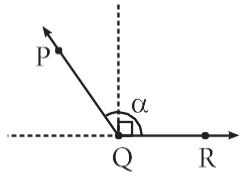


$0° < \beta < 90°$

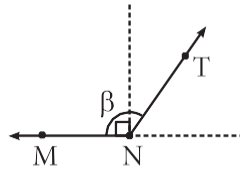


$0° < \theta < 90°$

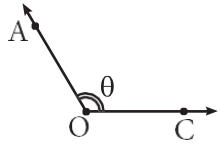
b) Ángulo obtuso



$90° < \alpha < 180°$

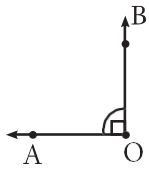


$90° < \beta < 180°$

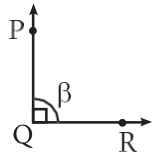


$90° < \theta < 180°$

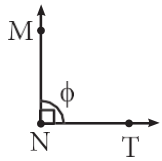
c) Ángulo recto



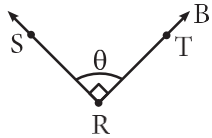
$\alpha = 90°$



$\beta = 90°$



$\phi = 90°$



$\theta = 90°$

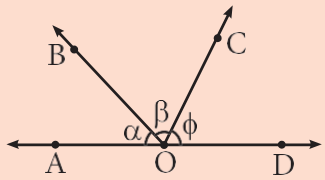
Veamos algunos ejemplos:

1. Ángulos agudos
10°, 30°, 60°, 80°, 89°, etc.

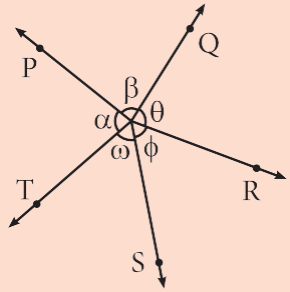
2. Ángulos obtusos
100°, 150°, 118°, 179°, 91°, etc.

• Recuerda que

Según el gráfico. se cumple que:



$\alpha + \beta + \phi = 180°$

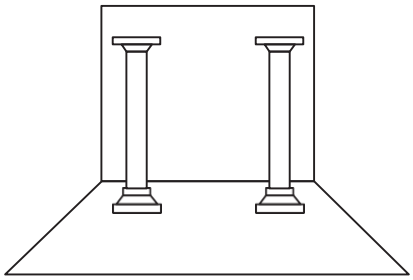


$\alpha + \beta + \theta + \phi + \omega = 360°$

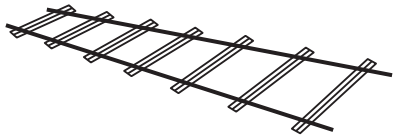
Ángulos en Rectas Paralelas y Secantes

• Marco teórico

Dos columnas de una casa nos dan la idea de rectas paralelas.

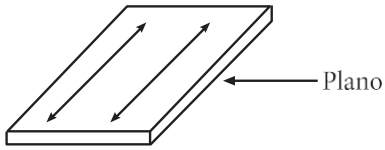


Los rieles del ferrocarril también nos da la idea de rectas paralelas.



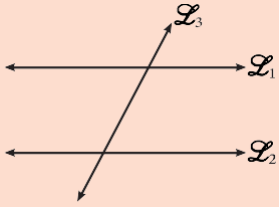
I. RECTAS PARALELAS

Son aquellas rectas que pertenecen a un mismo plano y que al prolongarlas no tendrán punto de contacto.

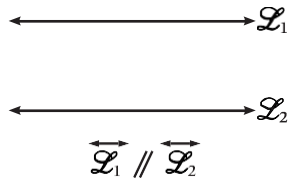


• Recuerda que

$\overleftrightarrow{L_3}$: Recta transversal o secante

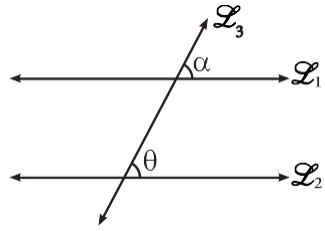


Notación:



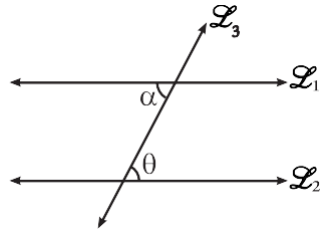
Notación: Se lee, $\overleftrightarrow{L_1}$ paralela a la $\overleftrightarrow{L_2}$.

1. Ángulos correspondientes



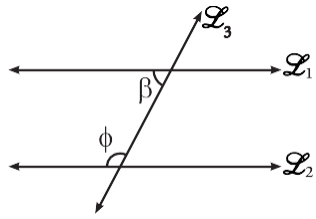
Si: $\overleftrightarrow{\mathcal{L}_1} \parallel \overleftrightarrow{\mathcal{L}_2} \Rightarrow \alpha = \theta$

2. Ángulos alternos internos



Si: $\overleftrightarrow{\mathcal{L}_1} \parallel \overleftrightarrow{\mathcal{L}_2} \Rightarrow \alpha = \theta$

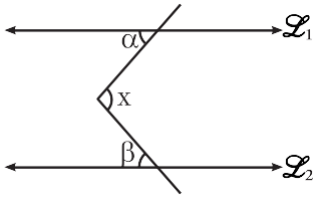
3. Ángulos conjugados internos



Si: $\overleftrightarrow{\mathcal{L}_1} \parallel \overleftrightarrow{\mathcal{L}_2} \Rightarrow \beta + \phi = 180°$

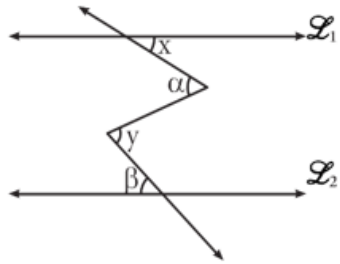
II. PROPIEDADES

1. Si: $\overleftrightarrow{\mathcal{L}_1} // \overleftrightarrow{\mathcal{L}_2}$



$$\boxed{x = \alpha + \beta}$$

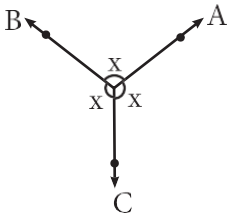
2. Si: $\overleftrightarrow{\mathcal{L}_1} // \overleftrightarrow{\mathcal{L}_2}$



$$\Rightarrow \boxed{x + y = \alpha + \beta}$$

• Ejercicios Resueltos

1. Calcular "x"



Resolución: Nos piden: "x"Del gráfico:

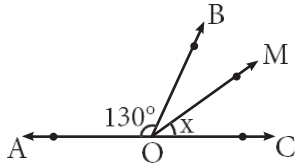
x + x+ x = 360°

Por tanto:

3x = 360°

x = 120°

2. Si OM es bisectriz del ángulo BOC, calcular "x"

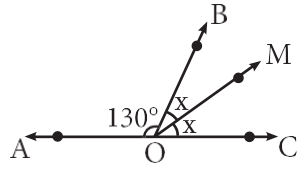


Resolución:

Nos piden: "x"

Como OM es bisectriz

$\Rightarrow m\measuredangle BOM = m\measuredangle MOC = x$

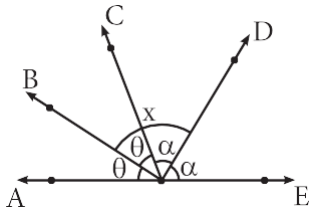


Luego: 130° + x + x =180°

2x = 50°

Por tanto: x = 25°

3. Calcular “x”



Resolución:

Nos piden: “x”

Del gráfico:

$$\theta + \alpha = x \ldots (1)$$

Además: $2\theta + 2\alpha = 180°$

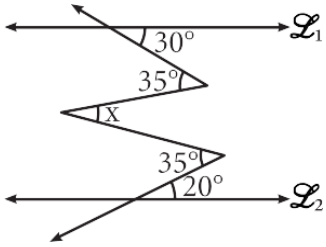
$$\theta + \alpha = 90° \ldots (2)$$

Reemplazando (1) en (2)

$$\theta + \alpha = 90°$$

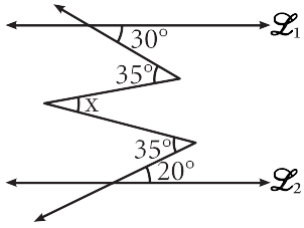
Por tanto: $x = 90°$

5. Calcula “x”, sí $\overleftrightarrow{\mathscr{L}_1} // \overleftrightarrow{\mathscr{L}_2}$.



Resolución:

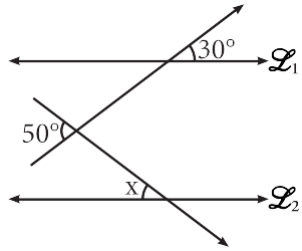
Nos piden: “x”.



En el gráfico, aplicamos la propiedad del serrucho.

$$35° + 35° = 30° + 20° + x$$
$$70° = 50° + x$$
$$20° = x$$

4. Si: $\overleftrightarrow{\mathscr{L}_1} // \overleftrightarrow{\mathscr{L}_2}$, calcula “x”.



Resolución:

Del gráfico, piden “x”.

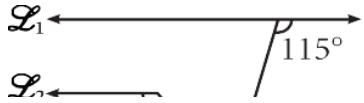
Aplicamos la propiedad:

$$x + 30° = 50°$$

Por tanto:

$$x = 20°$$

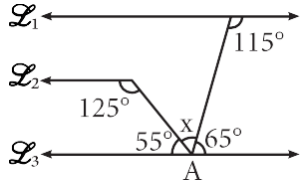
6. Calcula “x”, si $\overleftrightarrow{\mathscr{L}_1} // \overleftrightarrow{\mathscr{L}_2} // \overleftrightarrow{\mathscr{L}_3}$



Resolución:

Nos piden “x”.

En el gráfico, por ángulos conjugados



En el punto “A”.

$$x + 55° + 65° = 180°$$
$$x + 120° = 180°$$

Por tanto:

$$x = 60°$$